Learning Objectives

1. Determine how reaction rates compare by examining the factors contributing to effective collisions.

2. Interpret a Potential Energy (Reaction Coordinate Diagram) to determine the energy of activation for a reaction and whether the reaction is Exothermic or Endothermic.

3. Interpret the energy of the Activated Complex (Transition Energy) relative to the rate of reaction.

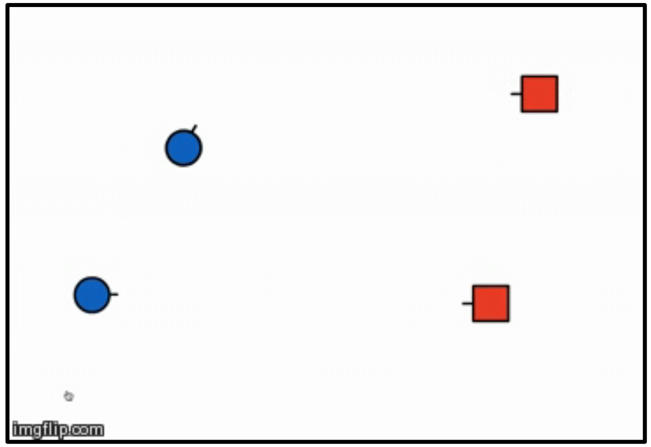
4. Determine how rate constants, k, compare by examining the relationships encompassed in the Arrhenius Equation, including frequency factor, E_a, and Temperature.

5. Correlate the magnitude of the frequency factor with the orientation of collisions.

Model of Chemical Kinetics: Collision Theory

aA + bB → cC + dD $\qquad r = k[A]^x[B]^y$

- Chemical reactions occur as a result of collision between reacting molecules.
- Higher Concentration, more chance for collisions.



Arrhenius Equation:

frequency factor (A): collision frequency
Unit: R

$$k = Ae^{-\frac{E_a}{RT}}$$

R: 8.413 J/(mol K)
T: in K

activation energy (E_a):
Unit: $\frac{J/kJ}{mol}$
a Path Function

Announcement 4/10

Assignments that are due soon

- Pre-Class Assignment Week 11-M

Office Hour

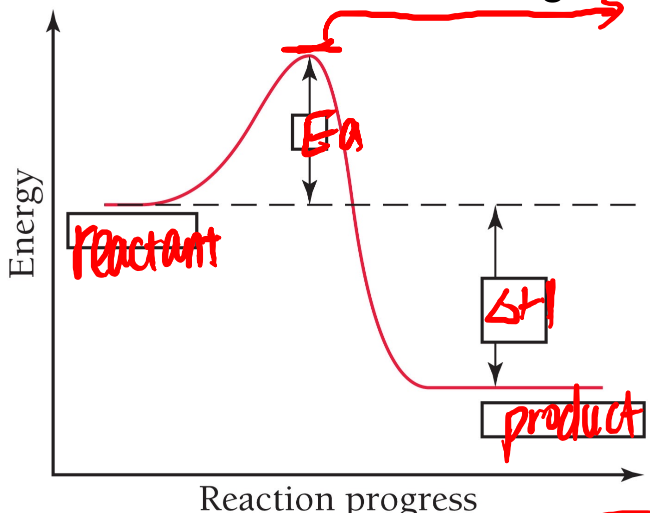
- Dr. Wang’s: Fri 2-3 PM at KC 382

Activation energy (E_a) = |transition − reactant|

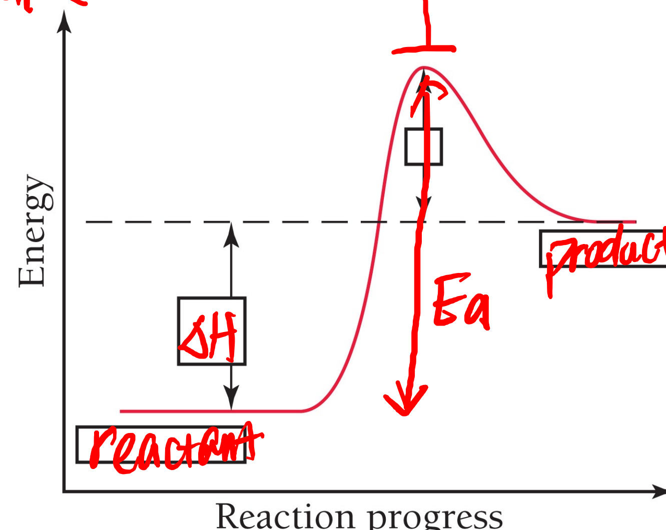
$$k = Ae^{-\frac{E_a}{RT}}$$

- Higher the activation energy, __slower__ the reaction rate.
- Reaction coordinate diagram:

transition state



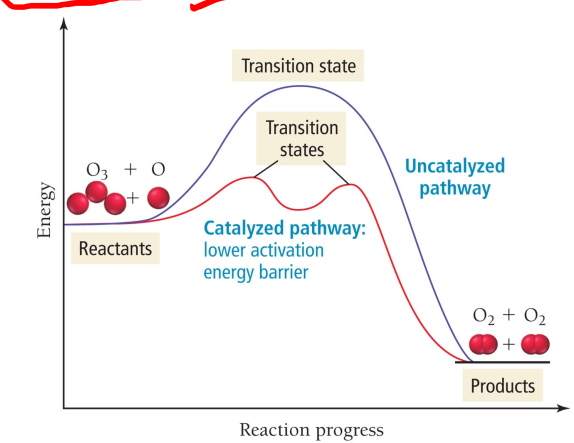
Endothermic / Exothermic



Endothermic / Exothermic

- Activation Energy for reversed reaction:

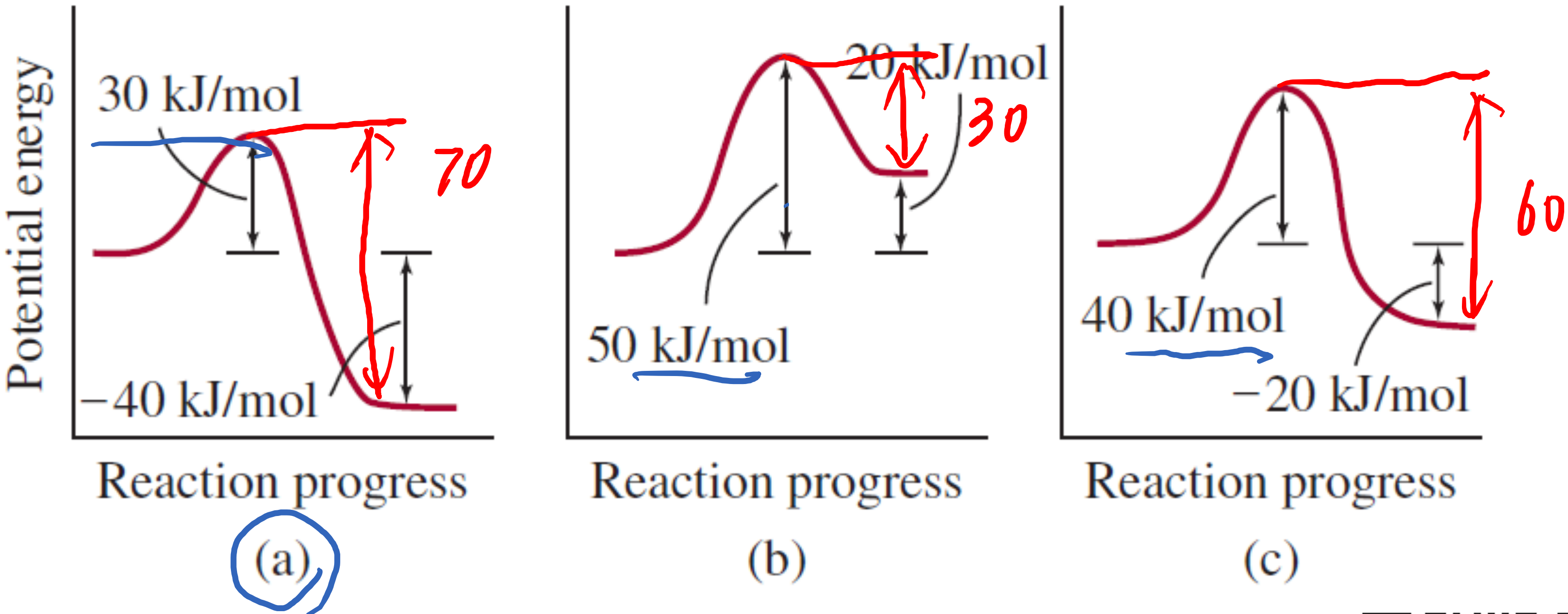
Ea = |transition − product|



Your Turn! $k = Ae^{-\frac{E_a}{RT}}$

Below are the reaction coordinate diagrams for three reactions and these three reactions all have the same order. Assume the three reactions have the same frequency factor (A), under the same temperature (T), which reaction is going to have the fattest rate?

s



Which reversed reaction is going to have the fastest rate? (b)

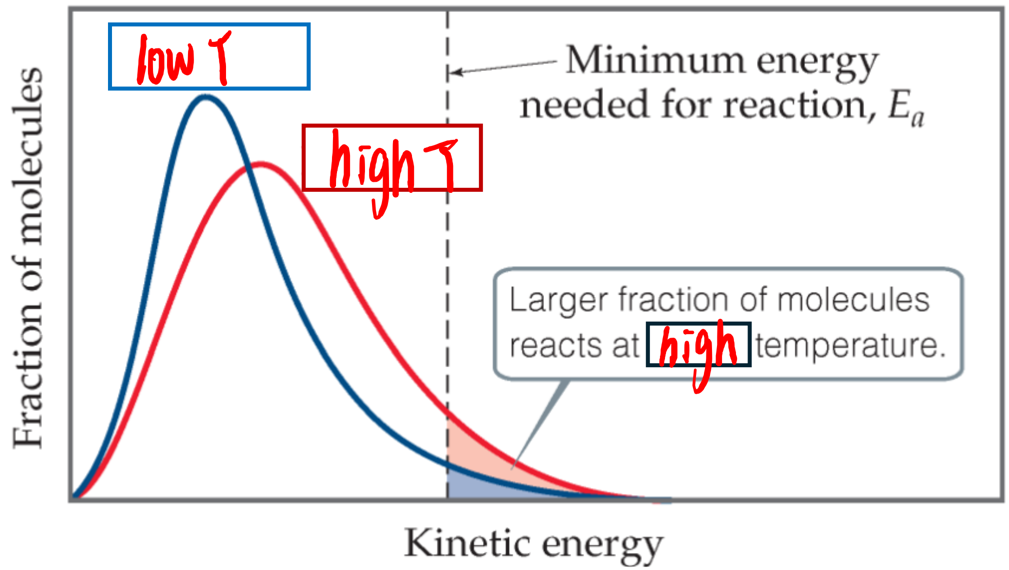
Temperature Dependance of Chemical Kinetics

$$k = Ae^{-\frac{E_a}{RT}}$$

- Higher temperature, __faster__ the reason rate.

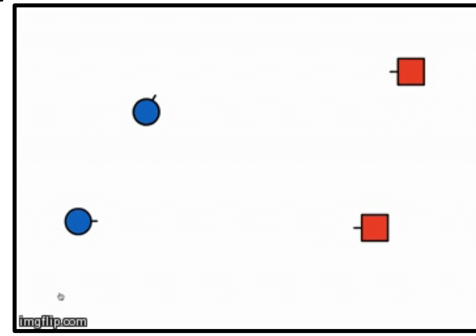
Fridge

Pressure Cooker



Frequency Factor (A)

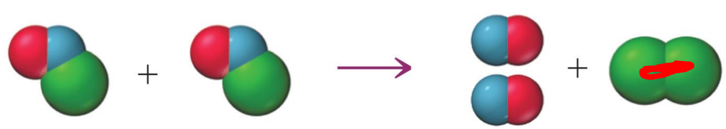
$$k = Ae^{-\frac{E_a}{RT}}$$



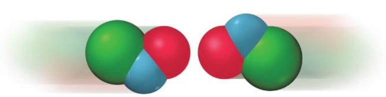
A = p×z

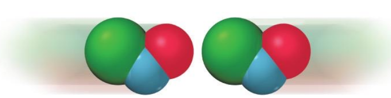
- z: collision frequency
- p: orientation factor (0 to 1)
  - p = 0.16 means:
    16 out of 100 collisions that have sufficient energy (higher than E_a) counts
  - complicated structures have a smaller p
  - p (atom) ≈ 1

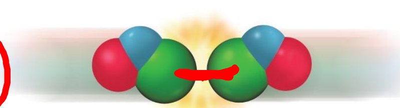
$NOCl(g) + NOCl(g) \longrightarrow 2\,NO(g) + Cl_2(g)$



Which collision is effective?

A. 

B. 

C. 

Which reaction would you expect to have the smallest orientation factor?

A. $H(g) + I(g) \rightarrow HI(g)$

I → H ← I
I → ← I

B. $H_2(g) + I_2(g) \rightarrow 2HI(g)$

H–H ⇔ I–I
I–I ⇔ H–H

C. $HCl(g) + HCl(g) \rightarrow H_2(g) + Cl_2(g)$ H–H   Cl–Cl

H–Cl ⇔ H–Cl X

H–Cl ⇔ Cl–H ✓

Your Turn! $k = Ae^{-\frac{E_a}{RT}}$

Reaction A + B → C + D has a rate law: Rate = k [A]. Energy of activation for this reaction is 106 kJ/mol (E_a) with a frequency factor of (A) $8.3 \times 10^{10}\ s^{-1}$ at 350K (T). Knowing that the rate of the reaction is 4.2×10^{-5} M/s at 100 s, what is the concentration of A at 100 s? R = 8.314 J/(mol K)

[A]

3.3 M

Your Turn! $k = Ae^{-\frac{E_a}{RT}}$

Reaction A + B → C + D has a rate law: Rate = k [A]. Energy of activation for this reaction is 106 kJ/mol with a frequency factor of 8.3 x 10^{10} s^{-1} at 350K. Knowing that the rate of the reaction is 4.2 x 10^{-5} M/s at 100 s, what is the concentration of A at 100 s? R = 8.314 J/(mol K)

$$k = 8.3\times10^{10}\,s^{-1}\times e^{-\frac{106\,\cancel{kJ}}{\cancel{mol}\times 8.314\frac{J}{\cancel{mol\cdot K}}\times 350\cancel{K}}\times\frac{10^3 J}{1\cancel{kJ}}} = 1.3\times10^{-5}\,s^{-1}$$

$$4.2\times10^{-5}\,\frac{M}{s} = 1.3\times10^{-5}\,s^{-1}\times[A]$$

$$[A] = 3.3\ M$$

$$[A] = [A]_0 e^{-kt}\quad X$$

Calculating E_a from Arrhenius Plot

- For a specific reaction, Ea and A are constant.
- Measuring k at different T

$$k = Ae^{-\frac{E_a}{RT}}$$

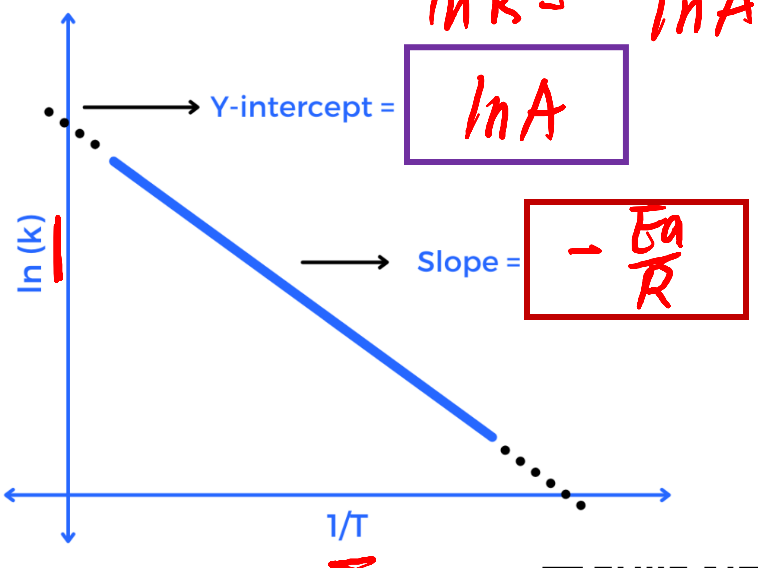
$$\ln k = \ln(A \times e^{-\frac{E_a}{RT}})$$

$$\ln k = \ln A + \ln e^{-\frac{E_a}{RT}}$$

$$\ln k = \ln A - \frac{E_a}{RT}$$

$$\underbrace{\ln k}_{y} = \left(\frac{-E_a}{R}\right)\underbrace{\left(\frac{1}{T}\right)}_{x} + \ln A$$

y = mx + b

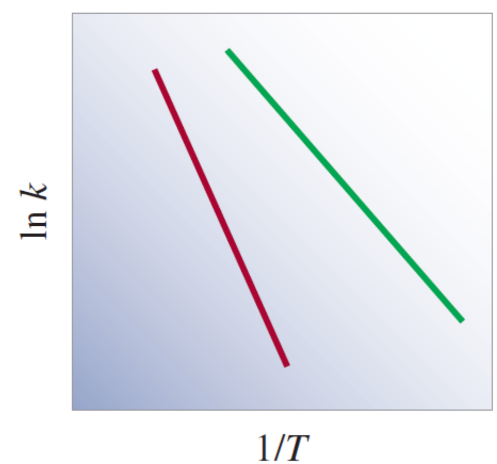


The diagram shows the plots of ln k versus 1/T for two first-order reactions. Which reaction has a greater activation energy?

A. Red

B. Green

C. Same



Calculating E_a from Arrhenius Plot

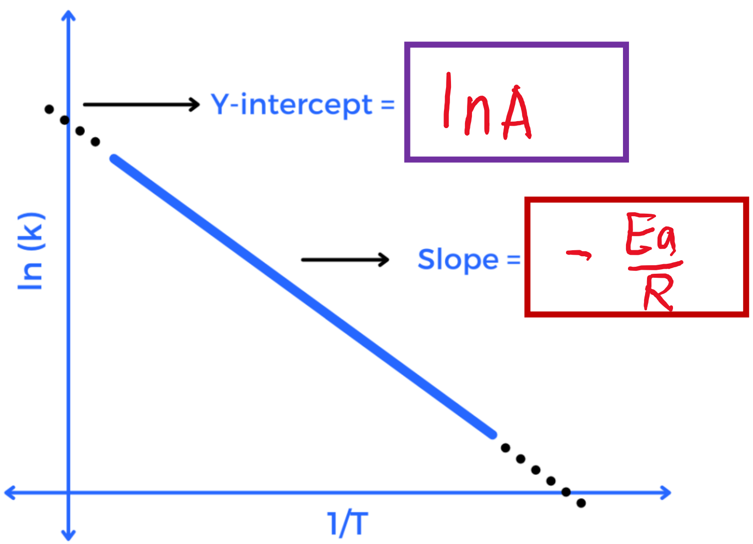
- For a specific reaction, __Ea__ and __A__ are constant.
- Measuring k at different __temperature__

$$k = Ae^{-\frac{E_a}{RT}}$$

$$\ln k = \ln\left(Ae^{-\frac{Ea}{RT}}\right) = \ln A - \frac{Ea}{RT}$$

$$\ln k = \left(\frac{-E_a}{R}\right)\left(\frac{1}{T}\right) + \ln A$$

y = mx + b



The diagram shows the plots of ln *k* versus 1/*T* for two first-order reactions. Which reaction has a greater activation energy?

A. Red

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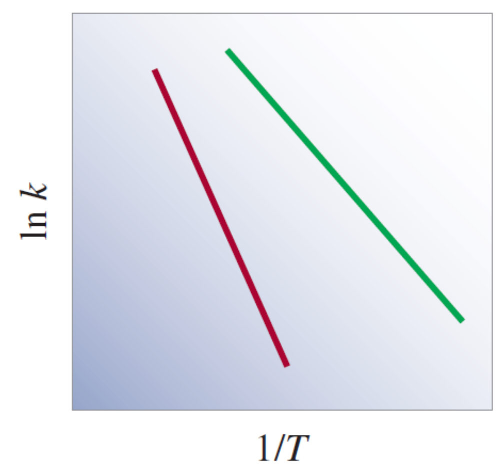


Your Turn! $\boldsymbol{lnk} = \left(\frac{-E_a}{R}\right)\left(\frac{1}{T}\right) + \boldsymbol{lnA}$

The results of a study for a first order reaction: $O_3(g) \rightarrow O_2(g) + O(g)$ has been analyzed and plotted into the graph below. Determine the value of the frequency factor and activation energy, in kJ/mol, for the reaction from the experimental results.

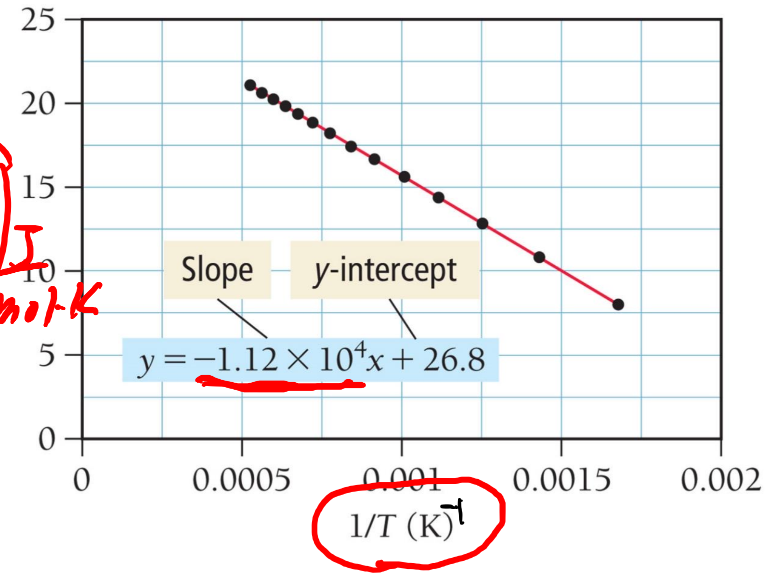
R = 8.314 J/(mol K)

$$-\frac{E_a}{R} = -1.12 \times 10^4 K$$

$$E_a = 1.12 \times 10^4 K \times 8.314 \frac{J}{mol \cdot K}$$

$$\ln A = 26.8$$

$$A = e^{26.8}$$



~~4.36*10^11 M-1s-1, 93.1 kJ/mol~~

Your Turn! $lnk = \left(\frac{-E_a}{R}\right)\left(\frac{1}{T}\right) + lnA$

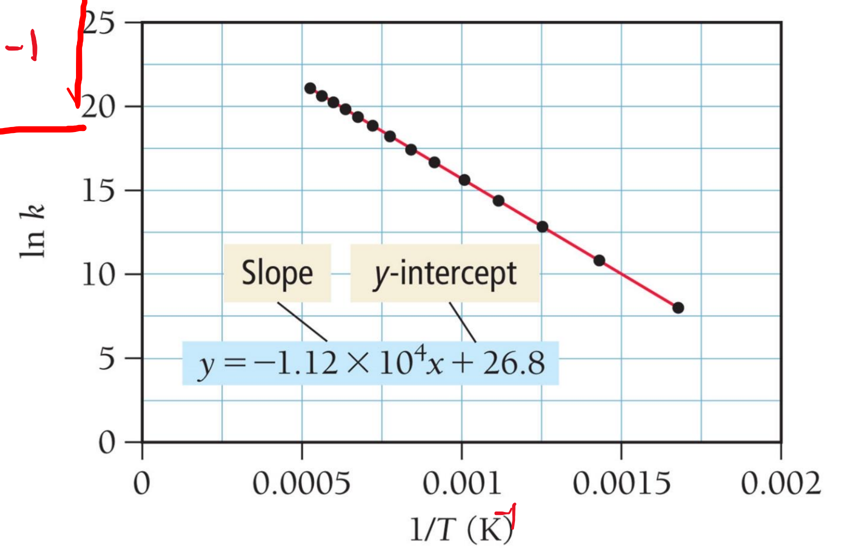
The results of a study for a first order reaction: $O_3(g) \rightarrow O_2(g) + O(g)$ has been analyzed and plotted into the graph below. Determine the value of the frequency factor and activation energy, in kJ/mol, for the reaction from the experimental results.

$\ln A = 26.8 \qquad A = e^{26.8}\,s^{-1} = \boxed{4.36 \times 10^{11}\,s^{-1}}$

$-\frac{E_a}{R} = \boxed{-1.12 \times 10^{4}\,K}$

$E_a = 1.12 \times 10^{4}\,K \times 8.314\,\frac{J}{mol \cdot K} \times \frac{1\,kJ}{10^3\,J}$

$= \boxed{93.1\,kJ/mol}$



Calculating E_a from Arrhenius Equation

- Only two data point (k1, T1; k2, T2) are measured.

$$lnk = \left(\frac{-E_a}{R}\right)\left(\frac{1}{T}\right) + lnA$$

Reaction 1: $\ln k_1 = \left(\frac{-Ea}{R}\right)\left(\frac{1}{T_1}\right) + \ln A$

Reaction 2: $\ln k_2 = \left(\frac{-Ea}{R}\right)\left(\frac{1}{T_2}\right) + \ln A$

① − ② $\ln k_1 - \ln k_2 = \left(-\frac{Ea}{R}\right) \times \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$\downarrow$

$$ln\frac{k_1}{k_2} = -\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Calculating E_a from Arrhenius Equation

- Only two data point (k1, T1; k2, T2) are measured.

$$\boldsymbol{lnk} = \left(\frac{-E_a}{R}\right)\left(\frac{1}{T}\right) + \boldsymbol{lnA}$$

① $\ln k_1 = -\frac{Ea}{R}\left(\frac{1}{T_1}\right) + \ln A$

② $\ln k_2 = -\frac{Ea}{R}\left(\frac{1}{T_2}\right) + \ln A$

① − ② $\ln k_1 - \ln k_2 = -\frac{Ea}{R}\left(\frac{1}{T_1}\right) - \left(-\frac{Ea}{R}\left(\frac{1}{T_2}\right)\right)$

$\ln \frac{k_1}{k_2} = -\frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$$\boldsymbol{ln}\frac{k_1}{k_2} = -\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Your Turn! $\boldsymbol{ln\frac{k_1}{k_2} = -\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$

The rate constant, $\boldsymbol{k}$, of a first-order reaction is $\mathbf{3.46{\times}10^{-2}}\ \boldsymbol{s^{-1}}$ at 298 K. What is the rate constant at 350 K if the activation energy, $\boldsymbol{E_a}$, for the reaction is 50.2 kJ/mol? R = 8.314 J/(mol K)

$\mathbf{0.702}\ \boldsymbol{s^{-1}}$

Your Turn! $\ln\frac{k_1}{k_2} = -\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

k_1 T_1

The rate constant, $\boldsymbol{k}$, of a first-order reaction is $3.46\times10^{-2}\ s^{-1}$ at 298 K. What is the rate constant k_2 at 350 T_2 K if the activation energy, E_a, for the reaction is 50.2 kJ/mol?

$$\ln\frac{3.46\times10^{-2}\,s^{-1}}{k_2} = -\frac{50.2\,\frac{\cancel{kJ}}{\cancel{mol}}\times\frac{10^3\,J}{1\,\cancel{kJ}}}{8.314\,\frac{J}{\cancel{mol}\cdot K}}\times\left(\frac{1}{298K} - \frac{1}{350K}\right)$$

$$e^{\left(\ln\frac{3.46\times10^{-2}\,s^{-1}}{k_2}\right)} = e^{(-3.01)}$$

$$\frac{3.46\times10^{-2}\,s^{-1}}{k_2} = 0.049$$

$$k_2 = 0.702\,s^{-1}$$

$0.702\ s^{-1}$

Announcements 9/30

Exam 2 at 11 AM on ~~10/7~~ 4/17:

- Due Date
- 50 minutes long
- Calculators (any kind), pens/pencils/highlighters, and erasers are allowed.
- Bonus point for Unit ~~2~~ 3: 3 Chem RePS worksheets and Exam ~~1~~ 2 Wrapper

Assignments that are due soon

- Pre-Class Assignment Week 11-W

Office Hour

- Dr. Wang's: Tue ~~10-11~~ 10:30-11:30 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue 12:00-1:00 PM at the TLC (Demille 160P)

Summary on Arrhenius Equation

$$k = Ae^{-\frac{E_a}{RT}}$$

- E_a: activation energy

  Larger E_a, slower the reaction.

- T: temperture

  Larger T, faster the reaction.

- A: frequency factor

  A depends on collision frequency and orientation factor

  ~0 asymetric complication

  1 atom

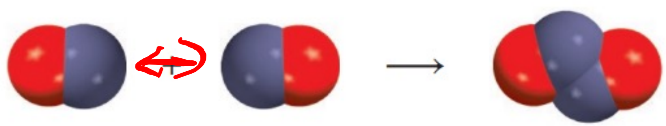
  Larger A, faster the reaction.

Elementary Steps and Reaction Mechanism

- Reaction Mechanism:

  the series of elementary steps that leads to product formation.
- Intermediate: Chemical which does not show up in the overall reaction.
- **Catalyst**: The intermediate which reacted with reactants and regenerated as a product in later steps

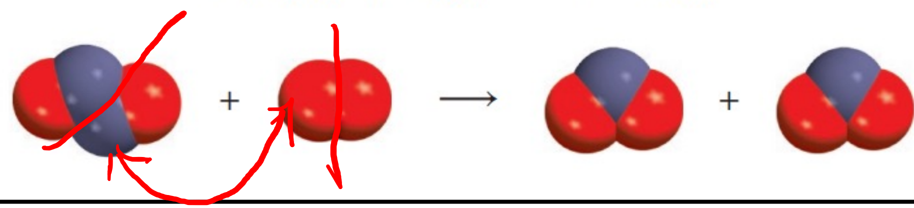
Step 1: $2NO(g) \longrightarrow \cancel{N_2O_2(g)}$



Intermediate: N_2O_2

It is a catalyst? $\boxed{X}$

Step 2: $\cancel{N_2O_2(g)} + O_2(g) \longrightarrow 2NO_2(g)$



Overall reaction: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

Your Turn!

A reaction following the elementary steps below. Which chemical is the intermediate for this reaction? Is it a catalyst? $\boxed{X}$

$$\textit{Step 1}\quad H_2(g) + ICl(g) \rightarrow \cancel{HI(g)} + HCl(g)$$

$$\textit{Step 2}\quad \cancel{HI}(g) + ICl(g) \rightarrow HCl(g) + I_2(g)$$

$$H_2(g) + 2ICl(g) \rightarrow 2HCl(g) + I_2(g)$$

A. H_2

B. ICl

C. HI

D. HCl

E. I_2

Your Turn!

The mechanism of a reaction is given below.

Step 1: $O_2 + \cancel{2NO} \rightarrow \cancel{2NO_2}$

Step 2: $\cancel{2NO_2} \rightarrow \cancel{2NO} + \cancel{2O}$

Step 3: $\cancel{2O} + 2O_2 \rightarrow 2\ O_3$

overall: $3O_2 \xrightarrow{NO} 2O_3$

Which chemical is the catalyst in this reaction?

first show up as reactant

A. O_2

B. NO

C. NO_2

~~D. NO~~

E. O

F. O_3

intermediate

Rate Law for Elementary Steps

- The molecularity of a reaction is the number of molecules reacting in **an elementary step**
- Only for elementary reactions can we deduce the rate law straight from the equation (coefficients = order)

TABLE 15.3 Rate Laws for Elementary Steps

Elementary Step	Molecularity	Rate Law
$A \longrightarrow$ products	$\boxed{1}$	Rate $= k\boxed{[A]}$
$A + A \longrightarrow$ products	$\boxed{2}$	Rate $= k\boxed{[A]}^2$
$A + B \longrightarrow$ products	$\boxed{2}$	Rate $= k\boxed{[A][B]}$
$A + A + A \longrightarrow$ products	$\boxed{3}$ (rare)	Rate $= k\boxed{[A]}^3$
$A + A + B \longrightarrow$ products	$\boxed{3}$ (rare)	Rate $= k\boxed{[A]^2[B]}$
$A + B + C \longrightarrow$ products	$\boxed{3}$ (rare)	Rate $= k\boxed{[A][B][C]}$

Your Turn!

The following reactions are all elementary step reactions.

Which following elementary step is a second order reaction?

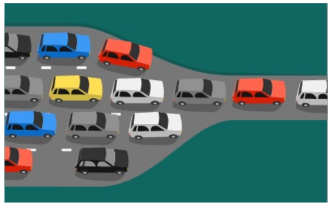
A. $X \longrightarrow$ products 1

B. $2X \longrightarrow$ products 2

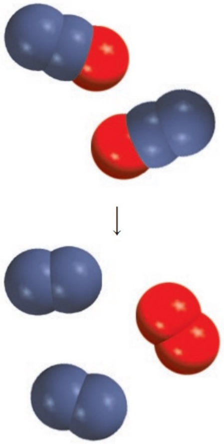
C. $X + Y + Z \longrightarrow$ products 3

D. $X + 2Y \longrightarrow$ products 3

Rate-Determining Step



- Rate-determining step: the slowest step in a sequence of elementary reactions that lead to the product(s)
- Typically, rate-determining step has the highest Ea
- Rate-determining step determines the rate of the overall reaction.
- Chem 150 focus on Rate-determining step = Step 1

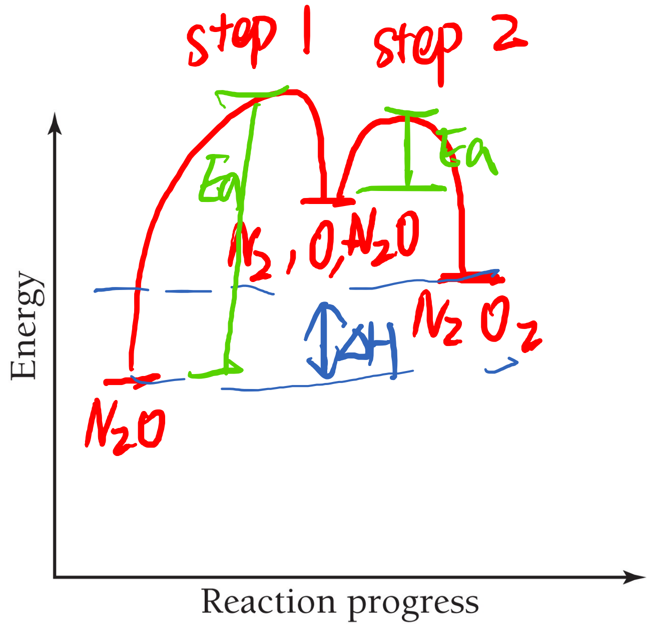


$$N_2O \xrightarrow{k_1} N_2 + \cancel{O}$$ step 1: slow (rate-determining step)

$$N_2O + \cancel{O} \xrightarrow{k_2} N_2 + O_2$$ step 2: fast

$$2N_2O \rightarrow 2N_2 + O_2$$ overall reaction

endothermic



Rate Law for the overall reaction: r = $k_1[N_2O]^1$

Order for the overall reaction: 1

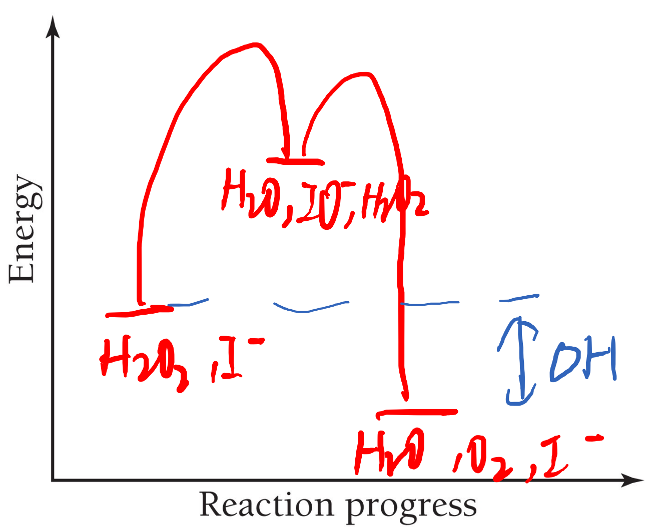
Your Turn!

Based on the information below, draft the reaction coordination diagram with reactants, intermediates, products and activation energy labeled. Write out the overall reaction and the rate law for the overall reaction.

exothermic rxn

Step 1: $H_2O_2 + \cancel{I^-} \xrightarrow{k_1} H_2O + \cancel{IO^-}$ step 1: slow (rate-determining step)

Step 2: $H_2O_2 + \cancel{IO^-} \xrightarrow{k_2} H_2O + O_2 + \cancel{I^-}$ step 2: fast



$$2H_2O_2 \rightarrow 2H_2O + O_2$$

$$r = k_1[H_2O_2][I^-]$$

What is the role of I^-? Catalyst